

ADI HADITYA NURSYAM

STEM & CODING EDUCATOR | SOFTWARE ENGINEERING

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EDUCATION & RECOGNITION

Bachelor of Engineering, Mechanical Engineering | Institut Teknologi Bandung **Aug 2022 - Jul 2026**

- GPA 3.91/4.00 | Mechanical Engineering Outstanding Student 2022 (one of three departmental students).

PROFESSIONAL SUMMARY

STEM and coding educator with 2+ years of tutoring experience across mathematics, physics, programming, and mechanical-engineering subjects, including nearly two years as a part-time Academic Tutor at Excelsis Learning Center (ELC). ITB Mechanical Engineering graduate (GPA 3.91/4.00) who brings applied software-engineering practice to teaching through JavaScript, TypeScript, Python, MATLAB, C/C++, full-stack systems, AI learning tools, and robotics.

TEACHING & TECHNICAL SKILLS

Teaching: Mathematics, Physics, Programming, Mechanical Engineering, concept explanation, structured problem solving, individualized guidance

Programming Languages: JavaScript, TypeScript, Python, MATLAB, C/C++

Software & Data: React, Next.js, Node.js, REST APIs, PostgreSQL, SQLite/Turso, Drizzle, Prisma, Git, testing

Engineering & Learning Systems: ESP32, PID, Web Serial, SOLIDWORKS, ANSYS Workbench, learning materials, question banks, assessments, progress tracking, AI/RAG

TEACHING EXPERIENCE

Founder & Lead Tutor / Full-Stack Engineer (CEO) | Zyx Academy (zyxacademy.com) **2024 - Present**

- Developed a private tutoring initiative into Zyx Academy; teach and support learners across mathematics, science, and mechanical-engineering subjects while leading learning operations.
- Built and operate the AI-enabled learning platform as primary engineer, serving 30+ learners with 100+ materials, question banks, assessments, progress tracking, and Zyra, a retrieval-backed AI tutor.

Academic Tutor (Part-Time) | Excelsis Learning Center (ELC) **Apr 2024 - Dec 2025**

Jatinangor, Indonesia | Hybrid

- Tutored university students in Mathematics, Physics, Programming, and Mechanical Engineering; prepared explanations and guided practice around individual learning needs.

Academic Tutor (Part-Time) | Mechanical Engineering ITB / Orpheus Engine **Aug 2023 - Aug 2026**

Bandung, Indonesia

- Provided peer academic support through concept review, structured problem solving, and one-on-one or group discussions; personally tutored 50+ learners across tutoring roles.

Academic & Practicum Assistant | Institut Teknologi Bandung **Jul 2023 - Jun 2024**

Bandung, Indonesia

- Guided students through algorithms, programming logic, structural mechanics, assessments, and technical problem solving in tutorials and practical sessions.

SOFTWARE ENGINEERING & APPLIED STEM

Head of IT Development / Full-Stack Engineer | HMM ITB **Aug 2024 - May 2026**

- Independently built and operated a typed learning and community platform unifying courses, resources, assessments, events, discussions, forms, content, and administration.
- Delivered authentication, role-based access, relational data, storage, notifications, PWA capabilities, and tests using Next.js, TypeScript, tRPC, PostgreSQL, Prisma, Auth.js, Zod, and Vitest.

Two-DOF SCARA Robot **2026**

- Built solo across SOLIDWORKS design, ESP32/C++ control, PID and feedforward, shared telemetry, Next.js/Web Serial HMI, and Python analysis as an applied robotics teaching platform.

Stack: ESP32, C++, Next.js, TypeScript, Python, Web Serial